
Envy-Free Chore Allocation with Unit Subsidies

under Negative Dichotomous Valuations

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Outline

- ① Setting and result
- ② Model
- ③ Failure of the goods construction
- ④ The construction
- ⑤ Pareto optimality
- ⑥ Summary

The setting

M : a set of m indivisible *chores*; $N = [n]$: agents, each with $v_i : 2^M \rightarrow \mathbb{R}$, $v_i(\emptyset) = 0$.

Each agent also receives a subsidy $p_i \geq 0$ from a third party.

Envy-free solution

$(\mathcal{A}, p)$ is an *envy-free solution* if

$$v_i(A_i) + p_i \geq v_i(A_j) + p_j \quad \text{for all } i, j.$$

Money enters linearly, at the same rate for everyone, and each item is worth at most one unit — the usual normalisation before absolute subsidy bounds mean anything.

The question is not existence. It is **how much money**.

Negative dichotomous valuations

For every agent i , every $S \subseteq M$ and every $e \notin S$,

$$v_i(S \cup \{e\}) - v_i(S) \in \{-1, 0\}.$$

The sign analogue of the dichotomous goods model. **Nothing further is assumed** — not additivity, not submodularity, not even subadditivity.

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Why binary marginals are natural for chores. An extra task either creates a fresh unit of burden, or is absorbed by a commitment already made:

- an agent already covering an on-call window absorbs another incident in that window at no extra cost;
- a driver already routed through a district absorbs one more stop.

Whether a chore is costly depends on the rest of the bundle — hence non-additivity is the point, not an artefact.

Main result

Theorem

For every instance with negative dichotomous valuations there exist a *complete* allocation $\mathcal{A}$ and a subsidy vector $p \in \{0, 1\}^n$ with

$$\sum_{i \in N} p_i \leq n - 1$$

such that $(\mathcal{A}, p)$ is an envy-free solution. Moreover $\mathcal{A}$ is *EF1 even before subsidies are paid*, and both are computable in polynomial time in the value-oracle model.

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- the per-agent bound is **0 or 1** — not merely $O(1)$;
- the total bound $n - 1$ is **tight**;
- EF1 is **extra** — it is not guaranteed by the goods-side construction.

What was already available here

Negative dichotomous valuations are a subclass of the *doubly monotone* valuations, so one bound comes for free.

Kawase, Makino, Sumita, Tamura, Yokoo (2025)

For doubly monotone valuations with marginals bounded by one in absolute value, any EF1 allocation converts into an envy-free solution with

$$n - 1 \text{ per agent,} \quad n(n - 1)/2 \text{ in total.}$$

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We sharpen $n - 1$ per agent to 1 , and $n(n - 1)/2$ to $n - 1$ in total: a linear-factor improvement in both.

The total bound $n - 1$ is tight

Take n agents, $n - 1$ chores, and identical *binary additive* costs:

$$v_i(S) = -|S| \quad \text{for every agent } i.$$

In any complete allocation some agent h receives the empty bundle. Envy-freeness of agent i towards h reads

$$-|A_i| + p_i \geq v_i(A_h) + p_h = p_h,$$

hence

$$p_i \geq |A_i| + p_h \geq |A_i|.$$

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Summing over all agents,

$$\sum_{i \in N} p_i \geq \sum_{i \in N} |A_i| = n - 1.$$

So no construction can do better in total.

□

Model and cost form

M : m indivisible chores. $N = [n]$: agents. $v_i : 2^M \rightarrow \mathbb{R}$ with $v_i(\emptyset) = 0$.

Algorithms run in the *value-oracle* model: an oracle returns $v_i(S)$ for a queried S .

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It is convenient to pass to non-negative *costs*

$$c_i(S) := -v_i(S).$$

Dichotomous cost function

$c(\emptyset) = 0$ and $c(S \cup \{e\}) - c(S) \in \{0, 1\}$ for all S and $e \notin S$.

Negative dichotomous valuations $\leftrightarrow$ dichotomous costs, under $c_i = -v_i$. In particular every c_i is **non-decreasing** and **integer-valued** — both used constantly below.

Write $c_i(e \mid S) := c_i(S \cup \{e\}) - c_i(S) \in \{0, 1\}$ for a marginal.

Fairness notions, in cost form

Envy-freeness

$c_i(A_i) \leq c_i(A_j)$ for all i, j .

Envy-free solution

$c_i(A_i) - p_i \leq c_i(A_j) - p_j$ for all i, j .

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EF1 for chores

A complete $\mathcal{A}$ is *EF1* if for all i, j either

$$c_i(A_i) \leq c_i(A_j), \quad \text{or} \quad \exists e \in A_i : c_i(A_i \setminus \{e\}) \leq c_i(A_j).$$

The removed item comes from the *envious agent's own* bundle.

Why not just mirror the goods construction?

The goods-side algorithm of Barman et al. is *incremental*: allocate items one at a time, maintaining the invariant that the partial allocation is envy-freeable with subsidies in $\{0, 1\}^n$.

The obvious chore analogue: give each new chore to an agent who currently receives *no* subsidy.

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This does not work.

There are negative dichotomous instances where a partial allocation admits an envy-free solution with subsidies in $\{0, 1\}^n$, yet after one more chore arrives *no* assignment of it preserves the unit-subsidy guarantee — even if the already-allocated bundles may first be permuted among the agents.

So the one-item extension principle fails for chores, and a different argument is needed. Here is the witness.

The witness instance

Three agents $N = \{1, 2, 3\}$, three chores $M = \{x, y, z\}$. Costs depend only on bundle size:

$$c_1(S) = \max \{0, |S| - 1\}, \quad c_2(S) = c_3(S) = \min \{|S|, 2\}.$$

$ S $	0	1	2	3
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Consider the partial allocation

$$X_1 = \{x, y\}, \quad X_2 = X_3 = \emptyset,$$

with z still unallocated.

The state is legitimately reachable

$$(\emptyset, \emptyset, \emptyset) \longrightarrow (\{x\}, \emptyset, \emptyset) \longrightarrow (\{x, y\}, \emptyset, \emptyset)$$

with pointwise-minimal subsidies

$$(0, 0, 0), \quad (0, 0, 0), \quad (1, 0, 0)$$

respectively — all within $\{0, 1\}^3$. So an incremental algorithm can legitimately arrive here.

At X : agent 1 costs her bundle at 1 and either empty bundle at 0, so she needs one unit; agents 2, 3 cost their own bundles at 0 and X_1 at 2, so they have no envy.

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Claim

z cannot be inserted so that the result admits an envy-free subsidy vector in $\{0, 1\}^3$ — even allowing $\{x, y\}, \emptyset, \emptyset$ to be permuted among the agents first.

The three ways to place z

(a) z joins the two-chore bundle Bundle sizes $(3, 0, 0)$. Every agent values a three-chore bundle at 2 and an empty bundle at 0, so whoever holds it needs a subsidy at least **two** units above an agent holding nothing. Impossible in $\{0, 1\}^3$.

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(b) z joins an empty bundle, and an agent of $\{2, 3\}$ holds the pair Bundle sizes $(2, 1, 0)$. That agent values her pair at 2 and the empty bundle at 0 — again a gap of **two**. Impossible.

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(b) z joins an empty bundle, and an agent of $\{2, 3\}$ holds the pair Bundle sizes $(2, 1, 0)$. That agent values her pair at 2 and the empty bundle at 0 — again a gap of **two**. Impossible.

(c) z joins an empty bundle, and agent 1 holds the pair The only remaining option. Let $i \in \{2, 3\}$ take the singleton and j the empty bundle. Then

$$\underbrace{1 - p_1 \leq 0 - p_i}_{\text{agent 1 vs } i} \implies p_1 \geq p_i + 1, \quad \underbrace{1 - p_i \leq -p_j}_{\text{agent } i \text{ vs } j} \implies p_i \geq p_j + 1.$$

Chaining: $p_1 \geq p_j + 2$ — impossible for $p \in \{0, 1\}^3$. □

The instance itself is fine

The obstruction is to the *method*, not to the instance.

The complete allocation $(\{x\}, \{y\}, \{z\})$ is envy-free **with no subsidy at all**.

Indeed agent 1 values every singleton at cost 0, and agents 2 and 3 value every singleton at cost 1 — nobody envies anybody.

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What went wrong. Reaching that allocation from $(\{x, y\}, \emptyset, \emptyset)$ requires *taking back* chores already placed on agent 1. An incremental construction that only ever adds items cannot do it.

This is why our proof starts from a *global* object — the terminal state of the Tao et al. algorithm — rather than building up one chore at a time.

The equality graph

A partial allocation X has *residue* $R(X) := M \setminus \bigcup_i X_i$ — the chores still unassigned.

Given X , define the *equality graph* $G = (N, E)$ on the agents:

$$(i, j) \in E \iff i \neq j \text{ and } c_i(X_i) = c_i(X_j).$$

An arc $i \rightarrow j$ records that agent i is *exactly indifferent* between her own bundle and j 's — she is on the verge of envying j .

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Strongly connected component; tail component

u, v are *strongly connected* if $u = v$, or each is reachable from the other. The equivalence classes are the *strongly connected components* (SCCs).

An SCC S is a *tail component* if no arc leaves it: there is no $(i, j) \in E$ with $i \in S$ and $j \notin S$.

Tail components exist

Lemma

Every finite non-empty directed graph has at least one tail strongly connected component.

PROOF. Start at any SCC. If it is not a tail component, follow an outgoing arc to another SCC and repeat.

No SCC can be visited twice: that would make the two components mutually reachable, hence a single component.

There are finitely many SCCs, so the walk terminates — at a component with no outgoing arc, i.e. a tail component. □

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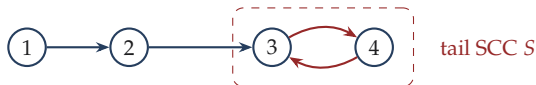
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Starting point: an envy-free *partial* allocation

Theorem [Tao, Wu, Yu, Zhou 2025, Thm. 6.1]

For every n and all dichotomous costs $c_1, \dots, c_n$ there is a polynomial-time algorithm returning a partial allocation X that is envy-free,

$$c_i(X_i) \leq c_i(X_j) \quad \text{for all } i, j,$$

and leaves $|R(X)| \leq n - 1$ chores unassigned.

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We need more than the statement — we need the **state the algorithm halts in**. So we recall its three rules.

It maintains X together with the equality graph G , and while $R(X) \neq \emptyset$ applies, *in order*:

The three rules

- (R1) If some $e \in R(X)$ and some i have $c_i(e \mid X_i) = 0$ — a *free* chore for i — assign e to i .
- (R2) Otherwise, if some $e \in R(X)$ and some arc $(i, j) \in E$ on a *directed cycle* of G have $c_i(e \mid X_j) = 0$, rotate the bundles along that cycle and assign e to i .
- (R3) Otherwise, select a tail SCC S of G . If $|R(X)| \geq |S|$, hand out distinct residual chores to the agents of S , one each. Otherwise **halt**.

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- (R3) Otherwise, select a tail SCC S of G . If $|R(X)| \geq |S|$, hand out distinct residual chores to the agents of S , one each. Otherwise **halt**.

So if the algorithm halts with residue left over:

- neither (R1) nor (R2) applies at that state, and
- for the tail SCC S selected at the halting step, $r := |R(X)| < |S|$.

That last inequality is what gives us *room*: more agents in S than chores left to place.

Proof of the theorem: (R1) and (R2)

Invariant: X is envy-free. It holds at the start, when every bundle is empty.

A sufficient condition. Suppose a step leaves every agent's own cost unchanged, and every new bundle X'_l contains some old bundle $X_{l'}$. Then envy-freeness survives: for every agent k ,

$$c_k(X'_k) = c_k(X_k) \leq c_k(X_{l'}) \leq c_k(X'_l). \quad (\spadesuit)$$

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(R2) Each agent on the cycle takes the bundle her arc points to. The arcs are equality arcs, so nobody's cost changes. Then e is free for i on X_j , which she now holds. Again $(\spadesuit)$ applies.

So the first two rules place a chore at **zero** cost to its recipient.

Proof of the theorem: (R3) and the bound

(R3) Each $i \in S$ receives one chore; (R1) does not apply, so her cost rises by **exactly one**. Agents outside S keep their costs and see bundles only grow. For $i \in S$ we need $c_i(X_i) + 1 \leq c_i(X'_j)$ for all $j \neq i$:

- $j \notin S$: S is a tail, so $(i, j) \notin E$, hence $c_i(X_i) < c_i(X_j) = c_i(X'_j)$; integrality gives $+1$.
- $j \in S$ and $c_i(X_i) < c_i(X_j)$: integrality, then monotonicity.
- $j \in S$ and $c_i(X_i) = c_i(X_j)$: then $(i, j) \in E$ lies inside an SCC, hence on a cycle. As (R2) does not apply, j 's new chore costs i exactly 1 on top of X_j .

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The bound Every iteration places at least one chore, so there are at most m . The loop ends with $R = \emptyset$, or halts in (R3) with $|R| < |S| \leq n$; either way $|R| \leq n - 1$. Each rule is testable in polynomial time, since an arc lies on a cycle iff its ends share an SCC. $\square$

The same two facts — S is a tail, and (R2) is inapplicable — are what our construction exploits at the halting state.

What the terminal state gives us

Fix X at termination; $R := R(X)$, $r := |R|$, G the equality graph there, and S the tail SCC from the halting step (so $r < |S|$).

Lemma (terminal state)

Suppose $r \geq 1$. Then

- (i) $c_i(e \mid X_i) = 1$ for every $i \in N$, $e \in R$;
- (ii) $c_i(e \mid X_j) = 1$ for every arc $(i, j) \in E$ with $i, j \in S$, and every $e \in R$;
- (iii) $c_i(X_i) < c_i(X_j)$ for every $i \in S$, $j \in N \setminus S$.

In words:

- (i) no leftover chore is free to anybody;
- (ii) inside S , no leftover chore is free even on a neighbour's bundle;
- (iii) agents in S are *strictly* better off than everyone outside S .

Proof of the terminal lemma

Since $r \geq 1$, the algorithm halted with non-empty residue. Hence neither (R1) nor (R2) was applicable.

(i) If $c_i(e \mid X_i) = 0$ for some i and $e \in R$, rule (R1) would apply. Marginals lie in $\{0, 1\}$, so $c_i(e \mid X_i) = 1$.

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- (ii) Let $(i, j) \in E$ with $i, j \in S$. As S is strongly connected there is a directed path $j \rightsquigarrow i$; take one without repeated vertices and append the arc (i, j) — a directed cycle through (i, j) .

If $c_i(e \mid X_j) = 0$ for some $e \in R$, rule (R2) would apply to that arc and that cycle. So the marginal is not 0, hence 1.

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(iii) Let $i \in S, j \notin S$. As S is a *tail* component no arc leaves it, so $(i, j) \notin E$, i.e.

$$c_i(X_i) \neq c_i(X_j).$$

Envy-freeness of X gives $c_i(X_i) \leq c_i(X_j)$. The inequality is therefore strict. □

The completion

Assume $r \geq 1$ and write $R = \{e_1, \dots, e_r\}$.

Since $r < |S|$, we may pick r *distinct* agents

$$T = \{t_1, \dots, t_r\} \subseteq S,$$

arbitrarily, and define

$$A_i := \begin{cases} X_{t_k} \cup \{e_k\}, & i = t_k \quad (1 \leq k \leq r), \\ X_i, & i \in N \setminus T. \end{cases}$$

The t_k are distinct and each chore of R is used once, so the bundles stay disjoint and now cover M : **$\mathcal{A}$ is complete.**

Agents in T are the *recipients* — each absorbs exactly one leftover chore, and by (i) that costs her exactly +1.

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What remains: decide *who gets paid*, and show one unit each is enough.

Who gets paid: the subsidy set

Definition (subsidy set)

Let $P \subseteq N$ be the *smallest* set with

(P1) $T \subseteq P$; and

(P2) if $i \in N \setminus S$ and $(i, j) \in E$ for some $j \in P$, then $i \in P$.

Then set $p_i := 1$ if $i \in P$, and $p_i := 0$ otherwise.

Equivalently: start from $P := T$ and repeatedly absorb any agent *outside* S with an equality arc *into* P — a **backward closure** in G .

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Let $P \subseteq N$ be the *smallest* set with

(P1) $T \subseteq P$; and

(P2) if $i \in N \setminus S$ and $(i, j) \in E$ for some $j \in P$, then $i \in P$.

Then set $p_i := 1$ if $i \in P$, and $p_i := 0$ otherwise.

Equivalently: start from $P := T$ and repeatedly absorb any agent *outside* S with an equality arc *into* P — a **backward closure** in G .

The intuition. A recipient's cost rose by one, so she needs a unit. Paying her may make her *neighbours* envious — exactly those who were indifferent to her, i.e. those with an equality arc into her. The payment must therefore propagate backwards.

The restriction $i \notin S$ in (P2) stops the closure from swallowing S itself — and that is what keeps the budget below n .

The algorithm

Step 2 is due to Tao et al.; steps 5–10 are the contribution

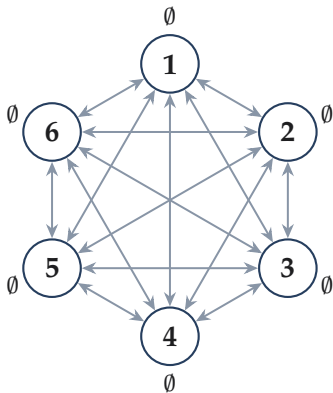
Input: agents $N = [n]$, chores M , value oracles $\{v_i\}$.

Output: complete $\mathcal{A}$ that is EF1, and $p \in \{0, 1\}^n$ with $(\mathcal{A}, p)$ envy-free, $\sum_i p_i \leq n - 1$.

1. Set $c_i = -v_i$ for every i .
2. Run the partial-allocation algorithm of Tao et al. on $\{c_i\}$, keeping the tail SCC selected at the terminating application of (R3) if the residue is non-empty.
3. Let X be its terminal partial allocation, $R = M \setminus \bigcup_i X_i$.
4. **if** $R = \emptyset$ **then return** $(X, 0)$.
5. Build the terminal equality graph $G = (N, E)$.
6. Write $R = \{e_1, \dots, e_r\}$; let S be that tail SCC. $r < |S|$
7. Choose distinct $T = \{t_1, \dots, t_r\} \subseteq S$.
8. $A_{t_k} \leftarrow X_{t_k} \cup \{e_k\}$ for $k \leq r$; $A_i \leftarrow X_i$ for $i \notin T$.
9. $P \leftarrow T$; **while** $\exists i \in N \setminus S, j \in P$ with $(i, j) \in E, i \notin P$: $P \leftarrow P \cup \{i\}$.
10. $p_i \leftarrow \mathbf{1}[i \in P]$; **return** $(\mathcal{A}, p)$.

The algorithm, run on an example

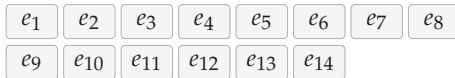
Start: every bundle is empty, so everyone is indifferent to everything — the equality graph is complete.



● (R1) free chore ● (R2) rotation ● (R3) tail SCC

Tao et al.'s rules

Leftover chores, $|R| = 14$



Reading the graph

An arc $i \rightarrow j$ means $c_i(X_i) = c_i(X_j)$: agent i is indifferent between her bundle and j 's.

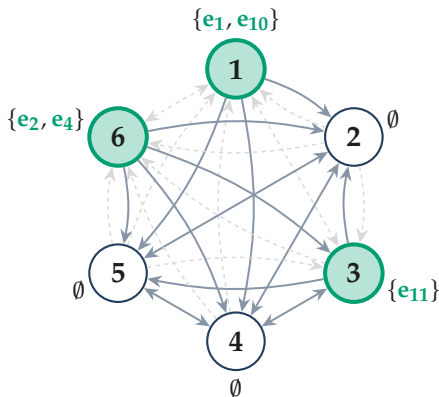
Every bundle is empty and $c_i(\emptyset) = 0$, so all 30 arcs are present.

Every step keeps X envy-free: $c_i(X_i) \leq c_i(X_j)$ for all i, j .

● final S ● T ● $P \setminus T$

The algorithm, run on an example

(R1) free chores placed: $e_1 \rightarrow 1$, $e_2 \rightarrow 6$, $e_4 \rightarrow 6$, $e_{10} \rightarrow 1$, $e_{11} \rightarrow 3$.



● (R1) free chore

● (R2) rotation

● (R3) tail SCC

● final S

● T

● $P \setminus T$

Tao et al.'s rules

Leftover chores, $|R| = 9$

e_3

e_5

e_6

e_7

e_8

e_9

e_{12}

e_{13}

e_{14}

Why (R1) applies

$$c_1(e_1 \mid \emptyset) = 0$$

$$c_6(e_2 \mid \emptyset) = 0$$

$$c_6(e_4 \mid \{e_2\}) = 0$$

$$c_1(e_{10} \mid \{e_1\}) = 0$$

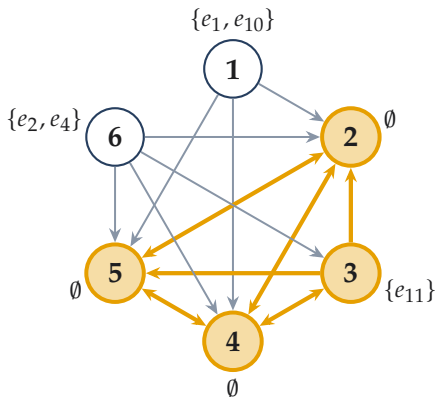
$$c_3(e_{11} \mid \emptyset) = 0$$

Now no leftover chore is free for anyone.

Dashed arcs were just lost: the bundle they point to grew, and now costs more.

The algorithm, run on an example

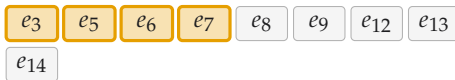
(R3) no free chore and no rotation: the tail SCC is $\{2, 3, 4, 5\}$, and $|R| = 9 \geq 4$.



● (R1) free chore
 ● (R2) rotation
 ● (R3) tail SCC

Tao et al.'s rules

Leftover chores, $|R| = 9$



Why (R3) applies

(R1) fails: each leftover chore costs every agent +1 on her own bundle.

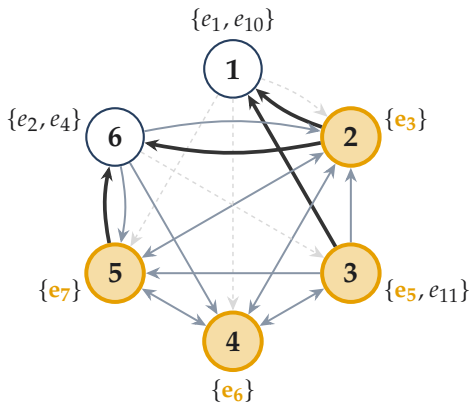
(R2) fails: $c_i(e \mid X_j) = 1$ for each leftover e and each arc $i \rightarrow j$ on a cycle.

No arc leaves $\{2, 3, 4, 5\}$: each member has cost 0, and every outside bundle costs each of them ≥ 1 .

● final S
 ● T
 ● $P \setminus T$

The algorithm, run on an example

(R3) one leftover chore to each member: $e_3 \rightarrow 2$, $e_5 \rightarrow 3$, $e_6 \rightarrow 4$, $e_7 \rightarrow 5$.



Tao et al.'s rules

Leftover chores, $|R| = 5$

e_8

e_9

e_{12}

e_{13}

e_{14}

Why no envy arises

Each member's cost rises by exactly one:

$$c_2(e_3 \mid \emptyset) = 1 \quad c_3(e_5 \mid \{e_{11}\}) = 1$$

$$c_4(e_6 \mid \emptyset) = 1 \quad c_5(e_7 \mid \emptyset) = 1$$

Outside bundles were already dearer, and (R2) fails inside the tail.

Bold arcs are new ties, e.g. $c_2(X_2) = c_2(X_1)$.

● (R1) free chore ● (R2) rotation ● (R3) tail SCC

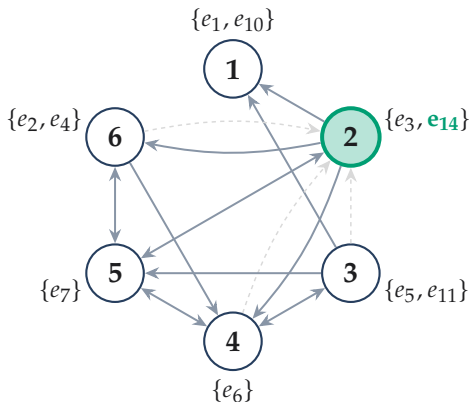
● final S

● T

● $P \setminus T$

The algorithm, run on an example

(R1) free chore placed: $e_{14} \rightarrow 2$.



Tao et al.'s rules

Leftover chores, $|R| = 4$

e_8

e_9

e_{12}

e_{13}

Why (R1) applies

$$c_2(e_{14} \mid \{e_3\}) = 0$$

Free only since (R3) gave her e_3 : before,
 $c_2(e_{14} \mid \emptyset) = 1$.

Now no leftover chore is free for anyone.

● (R1) free chore ● (R2) rotation ● (R3) tail SCC

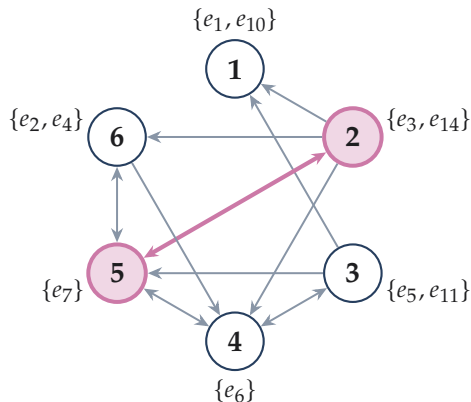
● final S

● T

● $P \setminus T$

The algorithm, run on an example

(R2) no free chore left, but the arc $2 \rightarrow 5$ lies on the cycle $2 \rightarrow 5 \rightarrow 2$.



Tao et al.'s rules

Leftover chores, $|R| = 4$

e_8

e_9

e_{12}

e_{13}

Why (R2) applies

(R1) fails: each leftover chore costs every agent +1 on her own bundle.

But $c_2(e_{12} \mid X_5) = 0$: e_{12} is free for 2 on top of 5's bundle,

and $2 \rightarrow 5$ lies on a cycle, so a rotation can hand 5's bundle to 2.

● (R1) free chore ● (R2) rotation ● (R3) tail SCC

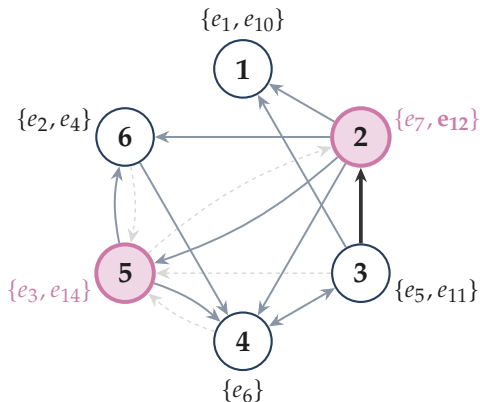
● final S

● T

● $P \setminus T$

The algorithm, run on an example

(R2) each agent on the cycle takes the bundle her arc points to; then agent 2 takes e_{12} .



● (R1) free chore
 ● (R2) rotation
 ● (R3) tail SCC

Tao et al.'s rules

Leftover chores, $|R| = 3$

e_8

e_9

e_{13}

Why no cost changes

Each arc of the cycle is an equality (X = the bundles before the move):

$$c_2(X_5) = c_2(X_2) = 1 \quad c_5(X_2) = c_5(X_5) = 1$$

so no cost on the cycle changes, and then
 $c_2(X_5 \cup \{e_{12}\}) = c_2(X_5)$.

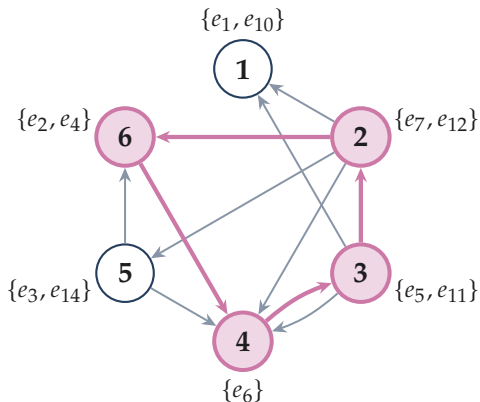
● final S

● T

● $P \setminus T$

The algorithm, run on an example

(R2) no free chore left, but the arc $3 \rightarrow 2$ lies on the cycle $3 \rightarrow 2 \rightarrow 6 \rightarrow 4 \rightarrow 3$.



Tao et al.'s rules

Leftover chores, $|R| = 3$

e_8

e_9

e_{13}

Why (R2) applies

(R1) fails: each leftover chore costs every agent +1 on her own bundle.

But $c_3(e_{13} \mid X_2) = 0$: e_{13} is free for 3 on top of 2's bundle,

and $3 \rightarrow 2$ lies on a cycle, so a rotation can hand 2's bundle to 3.

● (R1) free chore ● (R2) rotation ● (R3) tail SCC

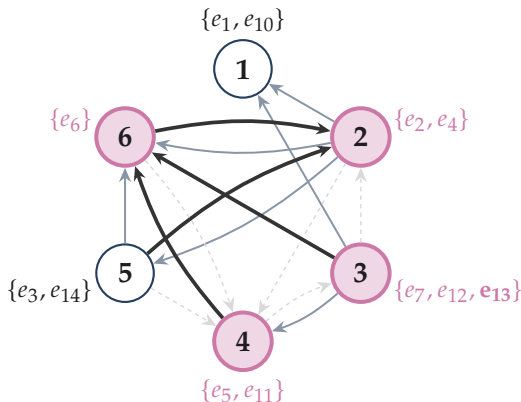
● final S

● T

● $P \setminus T$

The algorithm, run on an example

(R2) each agent on the cycle takes the bundle her arc points to; then agent 3 takes e_{13} .



Tao et al.'s rules

Leftover chores, $|R| = 2$

e_8

e_9

Why no cost changes

Each arc of the cycle is an equality (X = the bundles before the move):

$$\begin{aligned} c_3(X_2) = c_3(X_3) = 1 & \quad c_2(X_6) = c_2(X_2) = 1 \\ c_6(X_4) = c_6(X_6) = 0 & \quad c_4(X_3) = c_4(X_4) = 1 \end{aligned}$$

so no cost on the cycle changes, and then $c_3(X_2 \cup \{e_{13}\}) = c_3(X_2)$.

● (R1) free chore

● (R2) rotation

● (R3) tail SCC

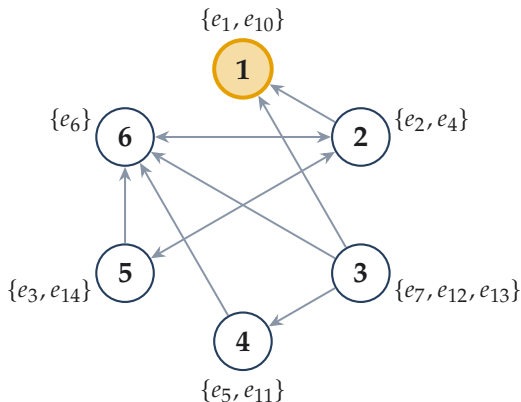
● final S

● T

● $P \setminus T$

The algorithm, run on an example

(R3) no free chore and no rotation: the tail SCC is $\{1\}$, and $|R| = 2 \geq 1$.



Tao et al.'s rules

Leftover chores, $|R| = 2$

e_8

e_9

Why (R3) applies

(R1) fails: each leftover chore costs every agent +1 on her own bundle.

(R2) fails: $c_i(e \mid X_j) = 1$ for each leftover e and each arc $i \rightarrow j$ on a cycle.

No arc leaves $\{1\}$: $c_1(X_1) = 0$, and every other bundle costs her ≥ 1 .

● (R1) free chore ● (R2) rotation ● (R3) tail SCC

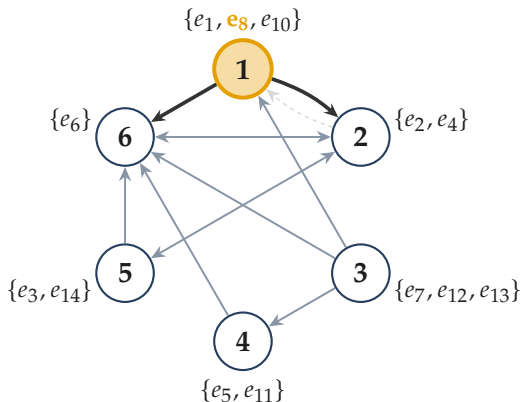
● final S

● T

● $P \setminus T$

The algorithm, run on an example

(R3) one leftover chore to each member: $e_8 \rightarrow 1$.



● (R1) free chore
 ● (R2) rotation
 ● (R3) tail SCC

Tao et al.'s rules

Leftover chores, $|R| = 1$

e_9

Why no envy arises

Her cost rises by exactly one:

$$c_1(e_8 \mid \{e_1, e_{10}\}) = 1$$

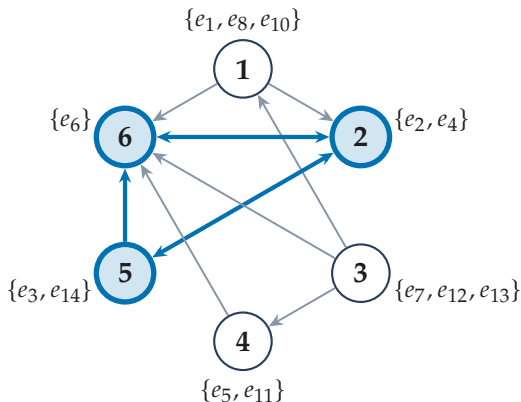
No envy: every other bundle was strictly dearer to her.

New arcs $1 \rightarrow 2, 1 \rightarrow 6$: her cost rose to $1 = c_1(X_2) = c_1(X_6)$.

● final S
 ● T
 ● $P \setminus T$

The algorithm, run on an example

Halt. The tail SCC $S = \{2, 5, 6\}$ has 3 agents, but only $r = 1$ chore left.



● (R1) free chore

● (R2) rotation

● (R3) tail SCC

● final S

● T

● $P \setminus T$

Tao et al.'s rules

Leftover chores, $|R| = 1$

e_9

Why the algorithm halts

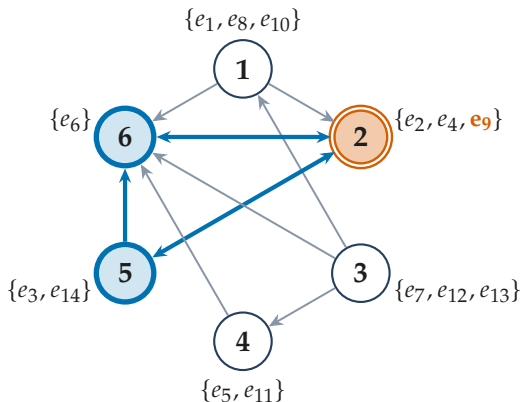
(R1), (R2) fail: e_9 costs everyone +1, on her own bundle and on every cycle arc.

(R3) needs one chore per agent of the tail $S = \{2, 5, 6\}$: 3 chores, but only $r = 1$.

Tao et al. stop here: X is envy-free, but e_9 is unassigned.

The algorithm, run on an example

Our completion. The leftover chore goes to distinct agents $T = \{2\} \subseteq S$: $e_9 \rightarrow 2$.



● (R1) free chore

● (R2) rotation

● (R3) tail SCC

● final S

● T

● $P \setminus T$

Our completion

Leftover chores, $|R| = 0$

none — every chore is placed

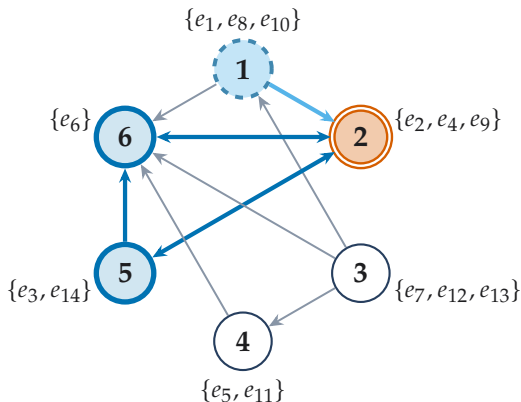
Why agent 2 must be paid

$c_2(e_9 \mid X_2) = 1$ by lemma (i), so
 $c_2(A_2) = 2 > 1 = c_2(A_5)$: unpaid, 2 would envy 5.

Inside S , by lemma (ii):
 $c_5(e_9 \mid X_2) = c_6(e_9 \mid X_2) = 1$, so a unit to 2 makes neither 5 nor 6 envious.

The algorithm, run on an example

Subsidy set P : agent 1, outside S , has an equality arc into P ($1 \rightarrow 2$), so she is paid.



Our completion

Leftover chores, $|R| = 0$

none — every chore is placed

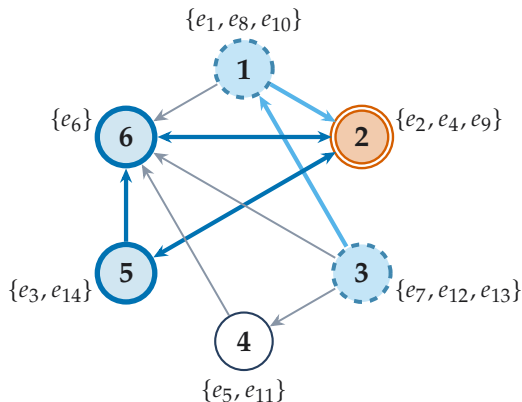
Why agent 1 is paid

Arc $1 \rightarrow 2$: $c_1(X_1) = c_1(X_2) = 1$, and e_9 is free for 1 on X_2 .

Once 2 is paid, $c_1(A_2) - p_2 = 0 < 1 = c_1(A_1)$: unpaid, 1 would envy 2.

The algorithm, run on an example

P grows again: agent 3, outside S , has an equality arc into P ($3 \rightarrow 1$), so she is paid.



Our completion

Leftover chores, $|R| = 0$

none — every chore is placed

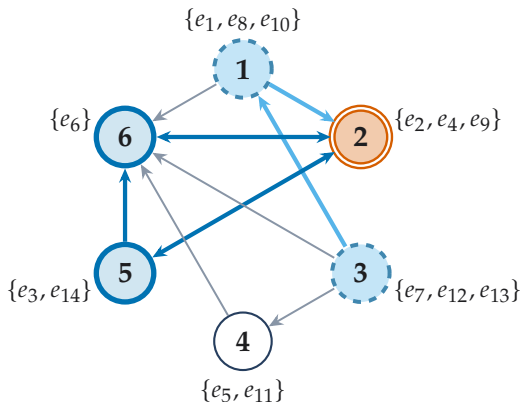
Why agent 3 is paid

Arc $3 \rightarrow 1$: $c_3(X_3) = c_3(X_1) = 1$, and $A_1 = X_1$.
Once 1 is paid, $c_3(A_1) - p_1 = 0 < 1 = c_3(A_3)$:
unpaid, 3 would envy 1.

No other agent outside S has an arc into P :
the closure stops.

The algorithm, run on an example

Result. One unit to each agent of P : envy-free, total $3 \leq n - 1 = 5$. Unpaid: $S \setminus T = \{5, 6\}$ and agent 4.



Our completion

Leftover chores, $|R| = 0$

none — every chore is placed

Why this is envy-free

$p = (1, 1, 1, 0, 0, 0)$, and $\sum_i p_i = 3 \leq n - 1 = 5$.

Agent 4: no arc into P , so

$c_4(A_4) = 1 \leq c_4(A_j) - 1$ for every $j \in P$.

Agents 5 and 6 in $S \setminus T$: lemma (ii) towards T ,
lemma (iii) towards $P \setminus T$.

Checked: $c_i(A_i) - p_i \leq c_i(A_j) - p_j$ for all i, j .

● (R1) free chore ● (R2) rotation ● (R3) tail SCC ● final S ● T ● $P \setminus T$

The closure does not eat into S

Lemma

$P \cap S = T$. Equivalently $P \setminus T \subseteq N \setminus S$: every subsidised agent inside S is a recipient.

PROOF. $T \subseteq P$ by (P1) and $T \subseteq S$, so $T \subseteq P \cap S$.

Suppose for contradiction some $i \in (P \cap S) \setminus T$, and put $P' := P \setminus \{i\}$.

P' still satisfies (P1), since $i \notin T$.

The closure does not eat into S

Lemma

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Suppose for contradiction some $i \in (P \cap S) \setminus T$, and put $P' := P \setminus \{i\}$.

P' still satisfies (P1), since $i \notin T$.

P' also satisfies (P2). Let $k \in N \setminus S$ with $(k, j) \in E$ for some $j \in P'$. Since $P' \subseteq P$ and P satisfies (P2), $k \in P$. Moreover $k \neq i$, because $k \notin S$ whereas $i \in S$. Hence $k \in P'$.

So P' satisfies both conditions and is strictly smaller than P — contradicting minimality. □

The budget

$$p \in \{0, 1\}^n \text{ and } \sum_{i \in N} p_i \leq n - 1.$$

PROOF. $p \in \{0, 1\}^n$ is immediate from the definition.

We have $|T| = r < |S|$, so $S \setminus T \neq \emptyset$.

By the previous lemma $P \cap S = T$, so every agent of $S \setminus T$ lies *outside* P and therefore receives **zero** subsidy.

At least one agent is unpaid, hence

$$\sum_{i \in N} p_i = |P| \leq n - 1.$$

□

The budget

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At least one agent is unpaid, hence

$$\sum_{i \in N} p_i = |P| \leq n - 1.$$

□

The slack came from $r < |S|$ — the halting condition of (R3).

The key lemma

Lemma

Let $i \in S$, $t \in T$ and $e \in R$. Then

$$c_i(X_t \cup \{e\}) \geq c_i(X_i) + 1.$$

That is: to an agent of S , *any* recipient's completed bundle looks at least one unit worse than her own.

PROOF. Envy-freeness of X gives $c_i(X_i) \leq c_i(X_t)$.

The key lemma

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PROOF. Envy-freeness of X gives $c_i(X_i) \leq c_i(X_t)$.

Strict case. If $c_i(X_i) < c_i(X_t)$, integrality gives $c_i(X_t) \geq c_i(X_i) + 1$, and monotonicity gives $c_i(X_t \cup \{e\}) \geq c_i(X_t)$. Done.

The key lemma

Lemma

Let $i \in S$, $t \in T$ and $e \in R$. Then

$$c_i(X_t \cup \{e\}) \geq c_i(X_i) + 1.$$

That is: to an agent of S , *any* recipient's completed bundle looks at least one unit worse than her own.

PROOF. Envy-freeness of X gives $c_i(X_i) \leq c_i(X_t)$.

Strict case. If $c_i(X_i) < c_i(X_t)$, integrality gives $c_i(X_t) \geq c_i(X_i) + 1$, and monotonicity gives $c_i(X_t \cup \{e\}) \geq c_i(X_t)$. Done.

Equality case. Suppose $c_i(X_i) = c_i(X_t)$.

- If $i = t$: terminal lemma (i) gives $c_i(e \mid X_i) = 1$.
- If $i \neq t$: then $(i, t) \in E$, and $i, t \in S$, so terminal lemma (ii) gives $c_i(e \mid X_t) = 1$.

Either way $c_i(X_t \cup \{e\}) = c_i(X_i) + 1$. □

Envy-freeness of the completion

Theorem

With $\mathcal{A}$ and p as constructed, $(\mathcal{A}, p)$ is an envy-free solution.

Fix i . Since $X_j \subseteq A_j$ for every j , monotonicity and envy-freeness of X give the standing inequality

$$c_i(A_j) \geq c_i(X_j) \geq c_i(X_i) \quad \text{for every } j. \quad (*)$$

We must check $c_i(A_i) - p_i \leq c_i(A_j) - p_j$ for every j . Three cases, according to where i sits:

$$\begin{array}{ccc} \underbrace{i \in T}_{\text{recipient}} & \underbrace{i \in P \setminus T}_{\text{paid, not a recipient}} & \underbrace{i \notin P}_{\text{unpaid}} \end{array}$$

(These are exhaustive: N is the disjoint union of T , $P \setminus T$ and $N \setminus P$.)

Case 1: $i \in T$ is a recipient

Then $p_i = 1$, and if $e \in R$ is the chore she absorbed, terminal lemma (i) gives

$$c_i(A_i) = c_i(X_i) + c_i(e \mid X_i) = c_i(X_i) + 1, \quad \text{so} \quad c_i(A_i) - p_i = c_i(X_i).$$

So we need $c_i(A_j) - p_j \geq c_i(X_i)$ for every j .

Case 1: $i \in T$ is a recipient

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So we need $c_i(A_j) - p_j \geq c_i(X_i)$ for every j .

$j \notin P$ Then $j \notin T$, so $A_j = X_j$ and $p_j = 0$. Envy-freeness of X :

$$c_i(A_j) - p_j = c_i(X_j) \geq c_i(X_i). \quad \checkmark$$

Case 1: $i \in T$ is a recipient

Then $p_i = 1$, and if $e \in R$ is the chore she absorbed, terminal lemma (i) gives

$$c_i(A_i) = c_i(X_i) + c_i(e \mid X_i) = c_i(X_i) + 1, \quad \text{so} \quad c_i(A_i) - p_i = c_i(X_i).$$

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$j \notin P$ Then $j \notin T$, so $A_j = X_j$ and $p_j = 0$. Envy-freeness of X :

$$c_i(A_j) - p_j = c_i(X_j) \geq c_i(X_i). \checkmark$$

$j \in P$ Here $p_j = 1$, so we need $c_i(A_j) \geq c_i(X_i) + 1$.

- If $j \in T$: this is exactly the key lemma.
- If $j \in P \setminus T$: then $j \notin S$ by the closure lemma, and $i \in S$, so terminal lemma (iii) gives $c_i(X_i) < c_i(X_j)$; integrality then gives $c_i(A_j) = c_i(X_j) \geq c_i(X_i) + 1$. $\checkmark$

Cases 2 and 3

Case 2: $i \in P \setminus T$ Here $A_i = X_i$ and $p_i = 1$, so $c_i(A_i) - p_i = c_i(X_i) - 1$.

For every j , the standing inequality (*) and $p_j \leq 1$ give

$$c_i(A_j) - p_j \geq c_i(X_i) - 1 = c_i(A_i) - p_i. \checkmark$$

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Case 3: $i \notin P$ Then $i \notin T$, so $A_i = X_i$, $p_i = 0$, and $c_i(A_i) - p_i = c_i(X_i)$.

If $j \notin P$: $A_j = X_j$, $p_j = 0$, and envy-freeness of X closes it.

If $j \in P$, we claim $c_i(A_j) \geq c_i(X_i) + 1$:

- $i \notin S$. Envy-freeness gives $c_i(X_i) \leq c_i(X_j)$. Equality would put $(i, j) \in E$; with $i \notin S$ and $j \in P$, rule (P2) would force $i \in P$ — contradiction. So the inequality is strict, and integrality with monotonicity gives the claim.
- $i \in S$. If $j \in T$ this is the key lemma. If $j \in P \setminus T$ then $j \notin S$ by the closure lemma, and terminal lemma (iii) plus integrality give the claim.

Since $p_j = 1$, we get $c_i(A_j) - p_j \geq c_i(X_i)$. $\checkmark$

□

EF1 comes for free

Corollary

$\mathcal{A}$ is EF1 — before any subsidy is paid. Precisely:

- (i) every non-recipient $i \in N \setminus T$ is *envy-free* in $\mathcal{A}$;
- (ii) every recipient t_k satisfies $c_{t_k}(A_{t_k} \setminus \{e_k\}) \leq c_{t_k}(A_j)$ for every j .

PROOF. Recall X is envy-free and $X_j \subseteq A_j$ for every j .

(i) For $i \notin T$ we have $A_i = X_i$, so for every j

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(ii) For $i = t_k$ we have $A_i = X_i \cup \{e_k\}$, so $A_i \setminus \{e_k\} = X_i$, and the same two steps give

$$c_i(A_i \setminus \{e_k\}) = c_i(X_i) \leq c_i(X_j) \leq c_i(A_j).$$

Removing the single chore she absorbed erases all her envy.

□

Putting it together

PROOF OF THE MAIN THEOREM. Pass to costs $c_i = -v_i$, dichotomous. Run the Tao et al. algorithm; let X be its terminal partial allocation, R the residue, $r = |R|$.

$r = 0$ X is already complete and envy-free, so $(X, \mathbf{0})$ is an envy-free solution with zero subsidy — and every envy-free allocation is EF1.

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$r \geq 1$ Build G , take the tail SCC S , and construct $T, \mathcal{A}, p$. Then

- $\mathcal{A}$ is complete;
- $(\mathcal{A}, p)$ is envy-free (envy-freeness theorem)
- $p \in \{0, 1\}^n, \sum_i p_i \leq n - 1$ (budget)
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Running time Tao et al. is polynomial. G has n vertices and is read off the n^2 values $c_i(X_j)$ — one oracle query each. SCCs, hence a tail component, in polynomial time. Completing $\mathcal{A}$ costs $r \leq n - 1$ assignments, and P is a backward reachability search in $O(n^2)$. $\square$

A further result on the domain

A separate question about negative dichotomous instances, independent of which algorithm produces the allocation: can one ask for the allocation to be *Pareto optimal* as well?

Pareto optimality is judged on the chore allocation alone, ignoring subsidies: $\mathcal{B}$ *dominates* $\mathcal{A}$ if $c_i(\mathcal{B}_i) \leq c_i(\mathcal{A}_i)$ for every i , strictly for some.

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Proposition

There are negative dichotomous instances — already with two agents and identical binary *submodular* costs — in which *no* Pareto-optimal allocation admits an envy-free subsidy vector in $\{0, 1\}^n$.

So on this domain, unit subsidies and Pareto optimality are simply incompatible — for *any* construction, not just ours.

A two-agent instance

Two agents, three chores $\{a, b, c\}$, both with

$$c_1(S) = c_2(S) = \min\{|S|, 2\}.$$

Marginals in $\{0, 1\}$, so dichotomous. Submodular, since $k \mapsto \min\{k, 2\}$ is non-decreasing and concave. Not additive: $c_i(\{a, b, c\}) = 2 < 3 = \sum_g c_i(\{g\})$.

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Up to symmetry every complete allocation has bundle sizes $(3, 0)$ or $(2, 1)$, giving cost profiles

$$(2, 0), \quad (2, 1), \quad (1, 2), \quad (0, 2).$$

The clash

The concentrated allocations are Pareto optimal None of $(2, 1)$, $(1, 2)$, $(0, 2)$ dominates $(2, 0)$ — each either raises some agent's cost, or matches it without improving the other. Symmetrically for $(0, 2)$.

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But they cannot be paid for If agent 1 holds all three chores, envy-freeness needs

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And the affordable allocations are dominated A split $(2, 1)$ has profile $(2, 1)$ and is envy-free with $(1, 0)$ — but handing the third chore to the agent already holding two moves the profile to $(2, 0)$: her cost is unchanged, the other's strictly falls. $\square$

Summary

Result

Negative dichotomous chores admit a complete allocation $\mathcal{A}$ and subsidies $p \in \{0, 1\}^n$ with $\sum_i p_i \leq n - 1$ such that $(\mathcal{A}, p)$ is envy-free; $\mathcal{A}$ is EF1 before subsidies; all computable in polynomial time.

The shape of the argument.

1. Terminal state of the Tao et al. algorithm: envy-free, $\leq n - 1$ chores left.
2. Its halting condition gives a tail SCC S with $r < |S|$ — *more agents than leftover chores*.
3. Park the leftovers on distinct $T \subseteq S$; each pays +1.
4. Pay T , propagating backwards along equality arcs, never into S .
5. $S \setminus T \neq \emptyset$ stays unpaid: exactly the $n - 1$ budget.

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5. $S \setminus T \neq \emptyset$ stays unpaid: exactly the $n - 1$ budget.

Also. The goods-side incremental construction does not transfer; and on this domain, Pareto optimality and unit subsidies are incompatible.

Directions

- **General binary marginals.** Marginals in $\{-1, 0, 1\}$ — goods and chores together, without additivity. Does the unit-subsidy guarantee survive? Open even for $n = 3$.
- **Beyond binary.** Marginals in $\{-k, \dots, 0\}$: what is the right per-agent bound?
- **Efficiency.** What is the least total cost achievable among allocations that admit a subsidy vector in $\{0, 1\}^n$?
- **Truthfulness.** Goko et al. obtain truthfulness for matroidal goods under free disposal. Is anything comparable possible for dichotomous chores?

Thank you

Questions welcome.

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